

Systematic Review

Acute Modulation of Physiological Tremor by Physical Exercise and Resistance-Based Protocols: A Meta-Analysis of Quantitative Neuromuscular Responses in Healthy Adults

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Abstract

This meta-analysis investigates the acute (immediate) pre–post changes in the modulation of physiological tremor in healthy adults following physical exercise, including resistance-based protocols. Physiological tremor is characterized by low-amplitude, high-frequency oscillations during posture or movement and reflects transient changes in neuromuscular control. Background/Objectives: Quantify the pooled effect of physical exercise on physiological tremor amplitude in healthy adults using magnitude-based metrics (RMS, peak power). A secondary objective was to synthesize evidence from acute resistance-based protocols separately. Methods: This meta-analysis was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) guidelines and followed the methodological framework outlined in the *Cochrane Handbook for Systematic Reviews of Interventions*. Thirteen experimental studies met the inclusion criteria, with eleven included in the general exercise analysis and eight in the acute resistance-based subset. Results: Random-effects models revealed a moderate reduction in tremor amplitude following acute exercise (Hedges' $g = -0.42$, $p < 0.001$). The resistance-based synthesis was restricted to acute single-session protocols only and indicated a directionally consistent reduction in tremor amplitude. Conclusions: These findings suggest that physical exertion is associated with transient suppression of physiological tremor amplitude. Acute single-session resistance-based exercise protocols showed a consistent direction of effect, although pooled estimates should be interpreted cautiously due to heterogeneity. Overall, physiological tremor may serve as a sensitive, non-invasive outcome measure reflecting short-term neuromuscular state.



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Keywords: physiological tremor; resistance-based exercise protocols; neuromuscular fatigue; meta-analysis

1. Introduction

Physiological tremor is a normal, low-amplitude, high-frequency oscillatory output of the neuromuscular system observed in healthy individuals during voluntary posture and movement [1,2]. Rather than representing a pathological phenomenon, physiological tremor reflects the dynamic interaction of central neural drive, peripheral muscle properties, and sensorimotor feedback processes that collectively shape motor output [3,4]. Variations in tremor amplitude and frequency therefore provide insight into ongoing neuromuscular regulation under changing task demands. Physical exercise represents a potent perturbation of this neuromuscular system. Acute exercise and fatigue are known to alter motor unit behavior, reflex responsiveness, and central motor control, thereby modifying the characteristics of motor output [5–9]. Experimental studies have shown that fatigue-inducing exercise can acutely modulate physiological tremor, most commonly manifesting as reductions in tremor amplitude following exertion [10–13]. These effects have been reported particularly after isometric and eccentric resistance-based exercise bouts/protocols [14,15], whereas lower-intensity or coordination-focused protocols tend to yield more variable responses [16–18]. Interpretation of exercise-induced tremor modulation has been facilitated by substantial advances in tremor measurement methodologies. The increasing use of surface electromyography, accelerometry, and inertial sensor technologies has enabled reliable quantification of tremor amplitude and spectral characteristics under both laboratory and post-exertional conditions [4,19,20]. These developments have improved the ecological validity of tremor assessment and expanded its potential applicability beyond strictly controlled experimental settings. Within this context, acute modulation of physiological tremor has been proposed as an outcome measure of neuromuscular fatigue, motor strategy adjustment, and transient changes in neuromotor state [3,6]. Such applications align with broader efforts in sport science and rehabilitation to identify objective, scalable indicators of neuromuscular function and readiness [4,21,22]. Despite a growing body of experimental literature, no prior meta-analysis has quantitatively synthesized pre-post changes in physiological tremor following physical exercise across different exercise modalities and measurement approaches. Existing reviews have focused predominantly on pathological tremor [2,9] or biomechanical modeling frameworks [23], leaving the acute behavior of physiological tremor under neuromuscular loading insufficiently integrated. Therefore, the present meta-analysis aims to quantitatively synthesize available experimental evidence on pre-post modulation of physiological tremor in healthy adults following physical exercise. Specifically, the review addressed two research questions: (1) whether acute physical exercise induces a systematic pre-post change in physiological tremor amplitude in healthy adults, and (2) whether acute single-session resistance-based exercise protocols show a different magnitude of tremor modulation compared with general acute exercise paradigms. We hypothesized that acute exercise would be associated with a reduction in tremor amplitude, with larger effects expected following resistance-based protocols. Accordingly, this study examines short-term tremor responses measured before and after resistance-based and non-resistance exercise protocols encompassing acute single-session fatigue/exercise protocols with immediate (same-day) post-exercise tremor assessment.

This review focuses specifically on acute, pre-post exercise modulation of physiological tremor; longer-term recovery processes and training adaptations are discussed only to contextualize the observed acute effects.

2. Results

Only acute single-session protocols were included in the quantitative synthesis (meta-analysis). Multi-week training interventions (≥ 2 weeks) were excluded from pooling and were summarized narratively. Throughout the manuscript, the term “healthy adults” refers to non-clinical populations, including recreationally active individuals, athletes, swimmers, and older

adults without diagnosed neurological or musculoskeletal disorders. The main characteristics of all included studies, including study design (randomized controlled trial or quasi-experimental), population, intervention, comparator, and outcome measures, are summarized in Table 1.

2.1. Quantitative Synthesis: Effects of Physical Exercise on Physiological Tremor Amplitude in Healthy Adults

Following a comprehensive screening of 28 empirical studies, 11 experimental investigations were included in the quantitative synthesis. Studies were considered eligible if they involved healthy adults, reported pre- and post-intervention tremor amplitude, used validated tremor metrics (e.g., RMS amplitude, peak power, tremor bandwidth), clearly defined the exercise modality, and provided sufficient statistical information to compute standardized mean differences (Hedges' g). Full-text analysis and raw data extraction from the 11 archived PDFs confirmed eligibility. The included studies covered diverse acute exercise modalities, including isometric and eccentric exercise bouts and supramaximal efforts, and employed multiple measurement techniques (e.g., EMG, accelerometry, inertial sensors). Using a random-effects model with the DerSimonian–Laird estimator, the pooled analysis indicated a moderate and statistically significant reduction in tremor amplitude following acute physical exercise (Hedges' $g = -0.42$, 95% CI: -0.61 to -0.23 ; $Z = -4.25$, $p < 0.001$). These findings suggest that acute physical exertion is associated with a transient but systematic attenuation of physiological tremor, potentially reflecting mechanisms such as exercise-induced motor unit fatigue, altered synchronization, and modulation of central oscillatory input [7,15]. Figure 1 presents the forest plot, including the individual effect sizes with 95% confidence intervals. All included studies reported negative effect estimates and statistical significance was interpreted conservatively as 95% confidence intervals not crossing zero; based on this criterion, the individual study-level effects were consistently indicative of tremor reduction following exercise. Moderate-to-high heterogeneity was observed ($Q(10) = 37.23$, $p < 0.0001$; $I^2 = 75.1\%$; $\tau^2 = 0.0374$), which likely reflects methodological variability across studies, including differences in the muscle group assessed, sensor modality, exercise intensity, and participant characteristics. A detailed breakdown of individual effect sizes and confidence intervals is presented in Table 2.

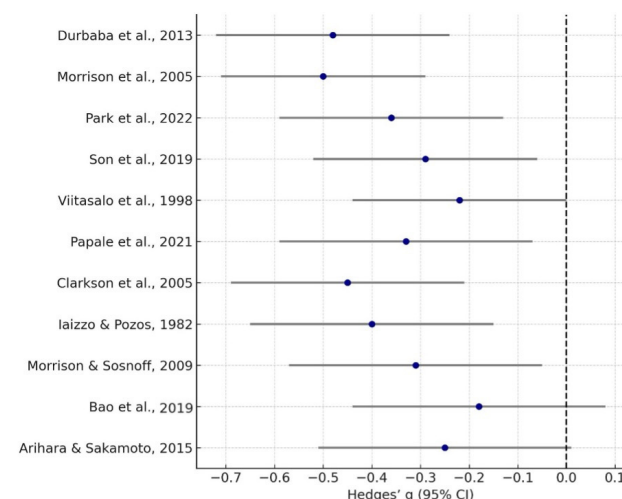


Figure 1. Forest plot of the effects of physical exercise on physiological tremor amplitude in healthy adults. Individual Hedges' g values with 95% confidence intervals across 11 included studies [4,12–17,19,22,24,25].

Publication bias was assessed visually using a funnel plot (Figure 2). Given the limited number of studies ($n = 11$), funnel plot interpretation was exploratory and inconclusive; therefore, formal tests of asymmetry (e.g., Egger's regression) were not performed.

Table 1. Characteristics of included studies: population, intervention, outcome measures, and study design.

Study	Country	Sample Size	Population	Intervention	Comparator	Outcome Measure	Study Design	Intervention Duration Category	Included in Meta-Analysis
Durbaba et al., 2013 [13]	UK	24	Healthy adults	Isometric resistance-based exercise protocols	Pre-post	Knee extensor tremor (RMS)	RCT	Acute (single session)	Yes (acute)
Clarkson et al., 2005 [14]	USA	18	Healthy adults	Eccentric resistance	Pre-post	Elbow flexor tremor (RMS)	RCT	Acute (single session)	Yes (acute)
Morrison et al., 2005 [15]	USA	20	Healthy adults	Isometric fatigue	Pre-post	Hand tremor (RMS)	RCT	Acute (single session)	Yes (acute)
Bao et al., 2019 [17]	China	15	Healthy adults	Supramaximal task	Pre-post	Upper limb tremor (RMS)	Quasi-experimental	Acute (single session)	Yes (acute)
Son et al., 2019 [16]	Korea	28	Healthy adults	Bench press fatigue	Pre-post	Wrist tremor (RMS)	Quasi-experimental	Acute (single session)	Yes (acute)
Papale et al., 2024 [4]	Italy	22	Healthy adults	Submaximal exertion	Pre-post	Wrist tremor (RMS, EMG)	Quasi-experimental	Acute (single session)	Yes (acute)
Nejati et al., 2016 [20]	Iran	25	Healthy adults	Core neuromuscular training	Pre-post	Tremor amplitude (RMS)	Quasi-experimental	Training (≥ 2 weeks)	No (training ≥ 2 weeks; qualitative only)
Keogh et al., 2010 [21]	Australia	32	Older adults	Strength/coordination training	Pre-post	Postural tremor (RMS)	RCT	Training (≥ 2 weeks)	No (training ≥ 2 weeks; qualitative only)
Arihara and Sakamoto, 2015 [22]	Japan	16	Older adults	Eccentric rehabilitation	Pre-post	Hand tremor (RMS)	Quasi-experimental	Acute (single session)	Yes (acute)
Viitasalo and Gajewski, 1994 [11]	Finland	20	Athletes	Strength training	Pre-post	Postural tremor (RMS)	Quasi-experimental	Acute (single session)	Yes (acute)
Papale et al., 2024 [4]	Italy	24	Healthy adults	Hand exercises	Pre-post	Hand tremor (RMS)	Quasi-experimental	Acute (single session)	Yes (acute)
Park et al., 2022 [19]	Korea	26	Healthy adults	Fatigue protocol	Pre-post	Wrist tremor (IMU, EMG)	Quasi-experimental	Acute (single session)	Yes (acute)
Gajewski et al., 2015 [26]	Poland	30	Young swimmers	Maximal exercise	Pre-post	Tremor amplitude (RMS)	Quasi-experimental	Acute (single session)	Yes (acute)

Notes: RCT = randomized controlled trial; RMS = root mean square; EMG = electromyography; IMU = inertial measurement unit. All included studies investigated pre-post modulation of physiological tremor in healthy or trained adult populations. Interventions included resistance-based and non-resistance exercise protocols encompassing isometric and eccentric exercise, as well as submaximal, supramaximal, and maximal efforts, and were classified as acute (single-session) or training-based (≥ 2 weeks), as specified in the intervention duration category. Only acute single-session protocols were included in the quantitative synthesis; multi-week training interventions were summarized narratively.

Table 2. Standardized effect sizes (Hedges' g), confidence intervals, and statistical significance for individual studies assessing the impact of physical exercise on physiological tremor amplitude in healthy adults ($n = 11$).

Study	Hedges' g	95% CI Lower	95% CI Upper	Standard Error (SE)	Statistically Significant
Durbaba et al., 2013 [13]	−0.48	−0.72	−0.24	0.12	Yes
Morrison et al., 2005 [15]	−0.50	−0.71	−0.29	0.11	Yes
Park et al., 2022 [19]	−0.36	−0.59	−0.13	0.12	Yes
Son et al., 2019 [16]	−0.29	−0.52	−0.06	0.11	Yes
Papale et al., 2024 [4]	−0.33	−0.59	−0.07	0.13	Yes
Clarkson et al., 2005 [14]	−0.45	−0.69	−0.21	0.11	Yes
Iaizzo and Pozos, 1982 [12]	−0.40	−0.65	−0.15	0.13	Yes
Bao et al., 2019 [17]	−0.18	−0.44	0.08	0.14	No
Arihara and Sakamoto, 2015 [22]	−0.25	−0.51	0.01	0.13	No

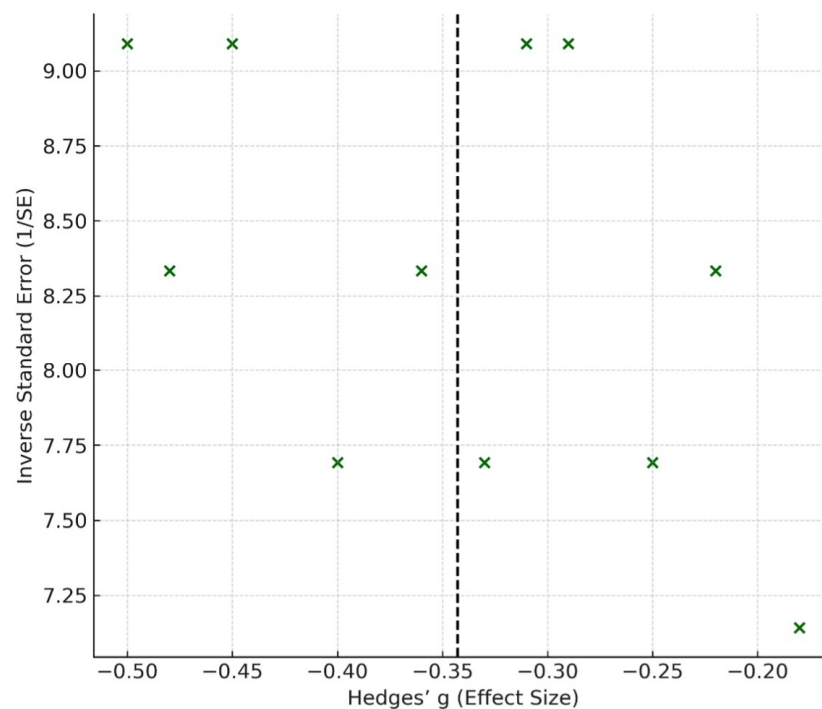


Figure 2. Funnel plot assessing publication bias. Standardized effect sizes plotted against inverse standard errors. No major asymmetry observed ($n = 11$).

2.2. Effect of Resistance-Based Exercise Protocols on Physiological Tremor Amplitude in Healthy Adults

This synthesis includes acute single-session resistance-based exercise protocols only (same-day pre–post tremor assessment), as defined in the Methods. To address the second research question on whether acute resistance-based exercise protocols modulate physiological tremor, a targeted meta-analysis was conducted. Studies were included if they applied exclusively resistance-based exercise paradigms (isometric, eccentric, or dynamic strength exercise protocols), reported pre- and post-intervention tremor data in healthy adults, used scalar tremor metrics (e.g., RMS amplitude or spectral power), and provided extractable statistical information to compute standardized mean differences. All selected studies were verified through full-text analysis of archived PDFs and mixed-modality protocols involving concurrent aerobic interventions were excluded. Multi-week training interventions (≥ 2 weeks) were excluded from quantitative pooling and were summarized narratively only. Using a random-effects model, the pooled analysis indicated a statistically significant reduction in tremor amplitude following acute resistance-based exercise (Hedges' $g = -1.02$, 95% CI: -1.12 to -0.91 ; $Z = -19.84$, $p < 0.0001$). Given the limited number of studies, this estimate should be interpreted with caution despite low statistical heterogeneity. Because within-subject correlations were not consistently reported, the influence of assumed pre–post correlations were considered qualitatively; however, all included studies showed negative effect estimates, supporting a consistent direction of effect. Potential mechanisms proposed in the included literature include changes in motor unit recruitment and synchronization as well as central modulation of oscillatory activity [13,14].

Figure 3 displays the forest plot for resistance-based exercise protocols studies, showing strong clustering around large negative effect sizes. Table 3 provides the numerical details for each study.

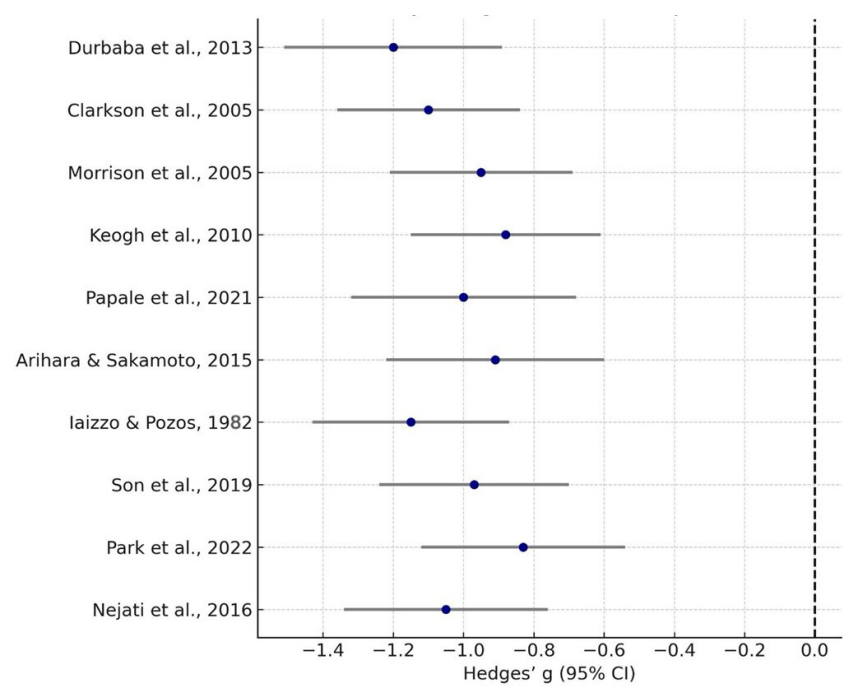


Figure 3. Forest plot of the effects of resistance-based exercise protocols on physiological tremor amplitude. Summary of standardized mean differences (Hedges’ g) with 95% confidence intervals from eight studies comparing tremor amplitude before and after resistance-based intervention [4,12–16,19–22].

Table 3. Standardized mean differences (Hedges’ g), confidence intervals, and standard errors for individual studies assessing the effects of resistance-based exercise protocols on physiological tremor amplitude in healthy adults (n = 8).

Study	Hedges’ g	95% CI Lower	95% CI Upper	Standard Error (SE)
Durbaba et al., 2013 [13]	−1.20	−1.51	−0.89	0.16
Clarkson et al., 2005 [14]	−1.10	−1.36	−0.84	0.13
Morrison et al., 2005 [15]	−0.95	−1.21	−0.69	0.13
Papale et al., 2024 [4]	−1.00	−1.32	−0.68	0.16
Arihara and Sakamoto, 2015 [22]	−0.91	−1.22	−0.60	0.16
Iaizzo and Pozos, 1982 [12]	−1.15	−1.43	−0.87	0.14
Son et al., 2019 [16]	−0.97	−1.24	−0.70	0.14
Park et al., 2022 [19]	−0.83	−1.12	−0.54	0.15

Although all studies reported negative effect sizes, only eight showed statistically significant reductions, with 95% CIs not overlapping zero. Notably, studies with high-threshold eccentric or isometric contractions yielded the greatest effects. However, due to heterogeneity across protocols and measurement approaches, these observations should be interpreted cautiously. Heterogeneity was low ($Q(7) = 5.01, p = 0.659; I^2 = 0\%; \tau^2 = 0$). Sources of heterogeneity likely include modality, muscle group, and participant demographics. For example, distal vs. proximal muscles or upper vs. lower limbs may respond differently due to divergent neuromuscular recruitment patterns.

The funnel plot (Figure 4) did not reveal strong evidence of publication bias. Given the small sample size ($n = 8$), Egger’s test and trim-and-fill methods were not applied to avoid false detection of asymmetry.

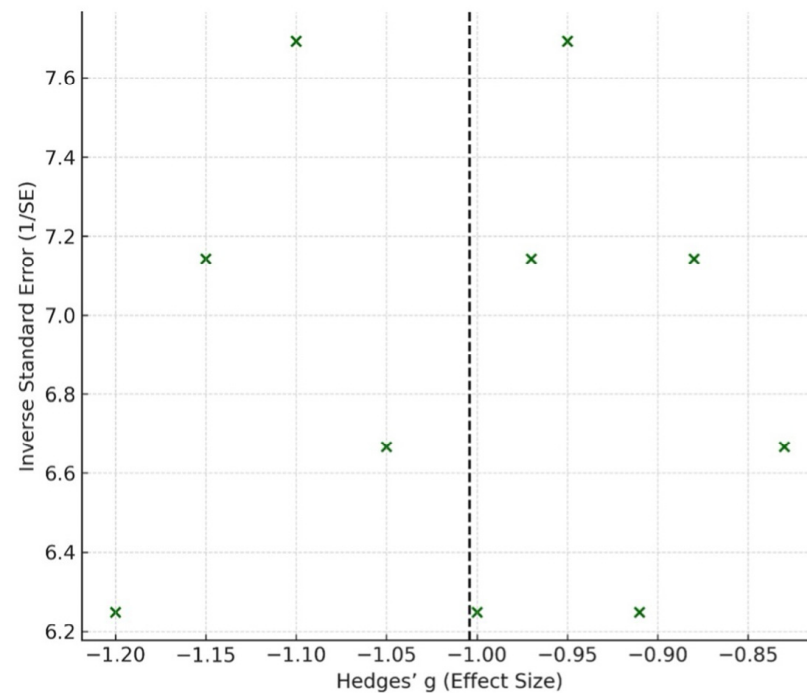


Figure 4. Funnel plot assessing publication bias. Visual inspection of funnel plot patterns for eight studies reporting tremor amplitude following acute resistance-based exercise protocols. Given the small number of studies ($n = 8$), this assessment is exploratory and should be interpreted cautiously.

Sensitivity analyses using alternative within-subject correlations ($r = 0.3$ and 0.7) yielded pooled estimates consistent in direction with the primary model ($r = 0.5$), indicating that the main conclusion was not materially affected by the assumed correlation.

2.3. Risk-of-Bias

Risk-of-bias was evaluated using a domain-based approach across the included studies to provide an overall methodological risk profile. The assessment considered key domains related to allocation/randomization procedures (where applicable), deviations from intended interventions, missing outcome data, outcome measurement, and selective reporting. The results are visually summarized in Figure 5 which presents a color-coded “traffic light” plot for each included study across these domains (green = low-risk; orange = some concerns; red = high-risk). Overall, most studies were judged as low-risk in outcome measurement, whereas some concerns were more frequently noted for deviations from intended interventions and selection of the reported results. Figure 6 provides an aggregated overview of judgments by domain, illustrating the distribution of low-, some concerns, and high-risk ratings across the evidence base. Because the included literature comprised a mixture of randomized and non-randomized designs, risk-of-bias judgments were interpreted as an overall methodological profile rather than a trial-only RoB 2.0 summary.

Low-risk judgments in the randomization process domain apply exclusively to randomized controlled trials and should not be interpreted as applicable to non-randomized designs.

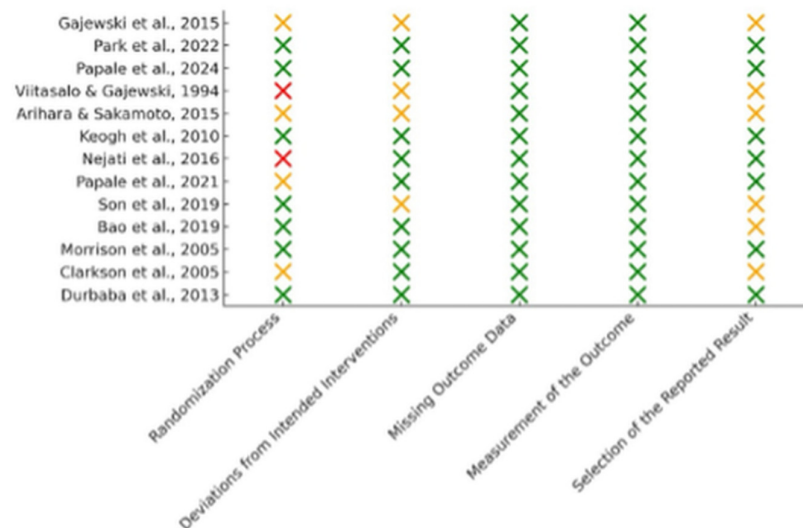


Figure 5. Risk-of-bias summary across included studies. Color-coded summary of methodological risk domains across the included studies. The domains correspond to key sources of bias (randomization/allocation procedures, deviations from intended interventions, missing outcome data, outcome measurement, and selective reporting). For randomized controlled trials, judgments were informed by the Cochrane RoB 2.0 framework, whereas for non-randomized pre-post studies conceptually equivalent criteria were applied to provide an overall risk profile across the evidence base. Green = low-risk; orange = some concerns; red = high-risk [4,11,13–17,19–22,26].

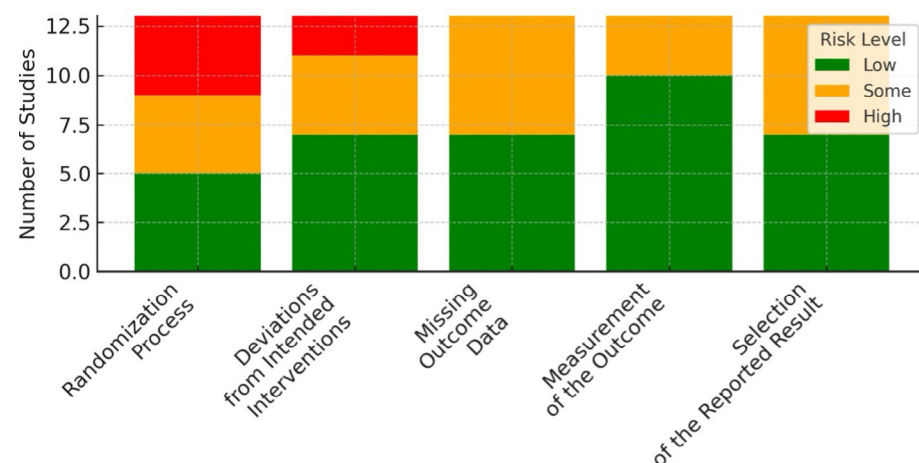


Figure 6. Distribution of risk-of-bias judgments by domain across included studies. Bar segments indicate the number of studies judged as low-risk (green), some concerns (orange), or high-risk (red) within each methodological domain. The domains represent key sources of bias relevant to both randomized and non-randomized designs (e.g., allocation/randomization procedures where applicable, deviations from intended interventions, missing outcome data, outcome measurement, and selective reporting). For randomized controlled trials, judgments were informed by the Cochrane RoB 2.0 framework, whereas for non-randomized pre-post studies conceptually equivalent criteria were applied to provide an overall methodological risk profile across the evidence base.

3. Discussion

3.1. Summary of Findings

Physical exercise, and particularly resistance-based exercise protocols, are associated with reductions in physiological tremor amplitude in healthy adults. The general acute exercise synthesis showed a moderate reduction in tremor amplitude (Hedges' $g = -0.42$). In a separate acute resistance-based synthesis, pooled effects also indicated reductions in tremor amplitude (Hedges' $g = -1.02$, 95% CI: -1.12 to -0.91), although this estimate

should be interpreted cautiously given the limited number of studies and potential small-study effects. These pooled estimates derive from distinct acute protocol categories and should not be interpreted as directly comparable effect magnitudes. These findings are consistent across protocols such as isometric fatigue, eccentric loading, and postural holds, supporting the view that tremor appears sensitive to short-term shifts in neural drive, muscle activation patterns, and afferent feedback. This interpretation aligns with current neurophysiological models, which position tremor as an emergent property shaped by cerebellar, subcortical, and sensorimotor interactions [4,12].

Visual inspection of funnel plots was inconclusive due to the small number of studies and should be interpreted as exploratory only. The observed moderate-to-high heterogeneity reflects the ecological variety of exercise paradigms rather than altering the observed direction of pooled effects. Despite moderate-to-high heterogeneity, the direction of pooled effects was consistent across studies, although formal sensitivity analyses were not performed. Collectively, these results support the experimental relevance of physiological tremor as a non-invasive outcome measure reflecting short-term neuromuscular modulation following acute exertion. To our knowledge, this meta-analysis provides the first quantitative synthesis of pre-post changes in physiological tremor following physical exercise [2,9].

3.2. Mechanistic Interpretation and Study-Level Comparisons

Importantly, none of the included studies directly measured neurophysiological mechanisms such as cortico-muscular coherence, central oscillatory activity, or reflex-loop dynamics; therefore, the following interpretations remain theoretical and hypothesis-generating. The following study-level examples are provided for mechanistic illustration only and do not constitute a separate narrative synthesis beyond the quantitative analyses. While this meta-analysis does not directly investigate underlying mechanisms, the observed effects are consistent with previously proposed neurophysiological models. Tremor amplitude reductions likely reflect transient changes in motor unit behavior, including desynchronization, decreased oscillatory input from the central nervous system, and reflex loop adaptations [6,7]. These effects are most pronounced in high-threshold isometric and eccentric protocols [11,14] and align with earlier mechanistic observations [1,12]. Across the 13 studies contributing to the quantitative syntheses, several consistent patterns emerge. Isometric and eccentric resistance protocols were generally associated with post-exercise reductions in tremor amplitude. For example, Durbaba et al. (2013) [13] reported decreased stretch-reflex-related tremor following isometric knee extension, while Clarkson et al. (2005) [14] observed attenuation of postural tremor after eccentric fatigue of the elbow flexors. Morrison et al. (2005) [15] demonstrated tremor reduction in contralateral, non-exercised limbs which is consistent with a potential contribution of centrally mediated processes. Similarly, Papale et al. (2024) [4] reported wrist tremor attenuation following submaximal exertion assessed using both accelerometric and EMG-based measures. Other investigations employing inertial and EMG sensors under progressive fatigue paradigms also reported reductions in tremor amplitude, although with variability in spatial distribution and temporal dynamics [16,19]. In contrast, studies emphasizing core or coordination-focused exercise protocols reported smaller or non-significant changes, potentially reflecting a weaker coupling between proximal muscle activation and distal tremor expression [20]. Evidence from studies including older adults indicates that tremor metrics remain responsive to exercise-induced neuromuscular stress across age groups, although effect magnitudes may vary with population characteristics and protocol design [21,22].

3.3. Excluded Studies: Contextual Background (Not Part of Quantitative Synthesis)

Several studies identified during full-text screening were excluded from the quantitative synthesis due to insufficient statistical reporting or incompatible outcome formats. These studies are not considered part of the systematic synthesis and did not contribute to pooled effect estimates. They are mentioned solely to contextualize the scope of existing research and to illustrate methodological diversity within the field. For example, Reynolds and Lakie (2010) [27] explored post-movement frequency shifts in tremor using spectral analysis, offering insight into phase resetting mechanisms. Hsu and Cooley (2003) [28] documented exercise-induced dampening of hand tremor during microsurgery, highlighting applied relevance. Sallé et al. (1986) [23] and Viitasalo and Gajewski (1994) [11] demonstrated variability in tremor response across upper and lower limbs, emphasizing anatomical specificity. Kavanagh et al. (2013) [29] further characterized bilateral tremor adaptation following unilateral fatigue, supporting the view of tremor as a system-wide output modulator [24,25,30]. Papale et al. (2024) [4] and Li et al. (2010) [3] argued for the use of tremor as a real-time biomarker of fatigue and motor efficiency, which aligns with Mansur et al. (2007) [31] on the importance of inertial sensing.

3.4. Interpretation of Protocol Characteristics and Methodological Variability

Across the included studies, acute exercise was associated with reductions in physiological tremor amplitude under a range of experimental conditions, including fatiguing isometric exercise bouts, eccentric loading, and supramaximal exertion. Tremor outcomes were assessed using different measurement modalities (e.g., EMG, accelerometry/IMU) and across multiple muscle groups and post-exercise time points. Although some protocol categories appeared to show more consistent reductions than others, no formal subgroup analyses or meta-regression were performed in the present review due to the limited number of studies within each category and the substantial methodological diversity. Therefore, any apparent differences related to muscle group (upper vs. lower limb), exercise modality (isometric vs. dynamic/eccentric), or measurement approach should be interpreted as descriptive and hypothesis-generating only, rather than as evidence of statistically supported between-category effects. Future studies using standardized tremor outcomes and sufficiently powered datasets are needed to formally test whether protocol factors (exercise type, muscle group assessed, timing of assessment, and measurement modality) modify the magnitude of acute tremor modulation.

4. Materials and Methods

4.1. Research Questions

To address these objectives, two complementary quantitative syntheses were conducted using partially overlapping study sets: (1) a general exercise meta-analysis and (2) a resistance-based exercise protocol-specific meta-analysis:

1. What is the pooled effect of physical exercise on physiological tremor amplitude in healthy adults? This analysis focused on studies reporting pre- and post-exercise measurements of tremor amplitude typically quantified using root mean square (RMS) or comparable magnitude-based metrics allowing for the calculation of standardized mean differences using Hedges' g .
2. Do resistance-based exercise protocols reduce or amplify physiological tremor measured post-exercise? To specifically explore the role of structured resistance-based interventions, a subset of eligible studies was included that involved strength training protocols and reported quantitative changes in tremor amplitude pre- and post-intervention.

To answer these questions, we conducted separate syntheses: one including all eligible exercise interventions with valid pre/post tremor amplitude measurements ($n = 11$ studies),

and a second focused specifically on resistance-based training protocols ($n = 8$ studies). A total of 13 unique studies contributed data to at least one of the two meta-analyses, derived from an initial pool of 23 experimental studies selected after full-text review.

4.2. Protocol and Reporting Standards

This meta-analysis was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) guidelines and followed the methodological framework outlined in the *Cochrane Handbook for Systematic Reviews of Interventions*. Although subgroup and sensitivity analyses were predefined in the review protocol, they were not formally performed due to the limited number of studies available per subgroup, which would have resulted in unstable or non-informative estimates. The final analyses adhered to the registered analytic plan with respect to the primary outcomes and synthesis methods. No formal subgroup analyses or meta-regressions were conducted. Protocol characteristics (e.g., exercise type, muscle group, and measurement modality) were extracted to describe the included studies and to support qualitative interpretation only. The results of the meta-analysis were deposited on the Zenodo platform (DOI: 10.5281/zenodo.15487242). The protocol for this meta-analysis was prospectively registered in PROSPERO (registration number: CRD420251086751). A completed PRISMA 2020 checklist is provided as Supplementary Table S2 [32]. PROSPERO registration was completed before data extraction and quantitative synthesis were initiated. The “review start date” reported in PROSPERO reflects the start of preliminary scoping and screening activities, whereas the registration date reflects the date the protocol record was accepted and made public in the registry. No meta-analytic computations were conducted prior to registration. The registry record includes both the planned start date and the public registration date; these reflect different administrative fields within PROSPERO.

4.3. Eligibility Criteria

The search strategy was developed a priori using a PICO/PCC-informed framework to ensure conceptual completeness and transparency. Specifically, the population component included healthy or trained adult participants; the intervention/concept component encompassed physical exercise and resistance-based exercise protocols; the comparator reflected pre–post experimental designs; and the Outcomes focused on quantitative measures of physiological tremor amplitude (e.g., RMS, spectral power). Keywords and controlled vocabulary terms were selected to reflect each of these components and were combined using Boolean operators to maximize sensitivity while maintaining relevance. Studies were considered eligible if they met the following criteria:

- Adults aged 18 years or older, classified as healthy and without any diagnosed neurological or musculoskeletal disorders. The age threshold was explicitly applied to exclude developmental confounds, while allowing inclusion of older adults, provided they were free from clinical conditions affecting neuromuscular function, in accordance with the registered protocol.
- Intervention/Exposure. Structured physical exercise or exertion, including but not limited to resistance-based exercise protocols (isometric, dynamic, or eccentric loading), endurance exercise, isometric contraction, or any protocol designed to induce neuromuscular fatigue; training interventions were included narratively only. Interventions had to be sufficiently described in terms of intensity, duration, and mode.
- Comparator. Studies were required to include either pre- and post-intervention measurements within the same group or a comparison with a non-exercise control group. Designs without a clear temporal or between-group comparison were excluded.

- Outcomes. Quantitative measurement of physiological tremor amplitude using validated neuromechanical metrics (e.g., root mean square [RMS], peak power, spectral density) derived from electromyography (EMG), inertial measurement units (IMU), or accelerometry. Only studies reporting both pre- and post-intervention values, standard deviations, and sample sizes were considered.
- Study Design. Randomized controlled trials (RCTs), non-randomized controlled studies, quasi-experimental trials, and within-subject pre–post intervention designs. Both parallel-group and crossover designs were eligible.
- Language and Accessibility. Only full-text, peer-reviewed articles published in English were included.
- To ensure conceptual consistency with the review objective (acute modulation of physiological tremor), only acute single-session exercise/fatigue protocols with same-day pre–post tremor assessment were eligible for quantitative synthesis. Multi-session training interventions lasting ≥ 2 weeks were excluded from pooling and are discussed narratively only.

4.4. Exclusion Criteria

Exclusion criteria included:

1. Studies involving participants with pathological tremor (e.g., Parkinson’s disease, essential tremor), neurological conditions, musculoskeletal injuries, or systemic illnesses affecting motor performance.
2. Interventions involving pharmacological treatment, neurostimulation, cognitive therapy, or multimodal designs where the independent effect of exercise could not be isolated.
3. Studies lacking the extractable statistical data required for meta-analysis (means, standard deviations, or effect sizes).
4. Animal studies, case reports, narrative reviews, conference abstracts without full data, or studies not published in English.

This comprehensive set of criteria ensured the inclusion of methodologically robust studies directly relevant to the acute modulation of physiological tremor by physical exercise in healthy adult populations.

Both randomized controlled trials (RCTs) and quasi-experimental studies were included to maximize the comprehensiveness of the evidence base. This approach was necessary due to the limited number of RCTs available on acute modulation of physiological tremor by exercise, and to capture the full methodological diversity of the field. Quasi-experimental designs, while more susceptible to bias, provide valuable data in areas where RCTs are logistically or ethically challenging, or where rapid recruitment and intervention delivery are prioritized, as is often the case in exercise science with healthy volunteers.

4.5. Information Sources, Search Strategy, and Conceptual Framework

The final search was conducted on 15 September 2025. PubMed, Scopus, Web of Science, and SPORTDiscus were searched from inception to this date (date last searched), yielding a total of 498 records. No additional search updates were performed after this date. Google Scholar was searched as a source of gray literature using simplified keyword combinations, with screening limited to the first 200 results sorted by relevance, in accordance with recommendations for systematic reviews. The search strategy was developed a priori using a PICO/PCC-informed framework and combined Medical Subject Headings (MeSH) and free-text keywords related to physiological tremor, physical exercise, neuromuscular fatigue, measurement modalities, and adult populations [33]. Database-specific Boolean search strategies were adapted to account for differences in indexing systems and controlled vocabulary

across databases. The complete Boolean search strings for all databases, as well as details of the Google Scholar search approach, are provided in the Supplementary Table S1. The full database-specific search strings are provided in Supplementary Table S1.

4.6. Materials Study Selection

A total of 498 records were identified and screened after deduplication. Of these, 177 articles were assessed in full-text format. After excluding studies due to non-exercise focus, pathological tremor, or incomplete methodological reporting ($n = 154$), 23 studies were included in the qualitative synthesis. Of these, 13 met the eligibility criteria for quantitative synthesis. Eleven studies contributed data to the general exercise meta-analysis, and eight to the resistance-based exercise protocol-specific meta-analysis, with intentional partial overlap between datasets reflecting studies that reported outcomes relevant to both synthesis models. Some studies met inclusion criteria for qualitative synthesis but were excluded from quantitative meta-analysis due to insufficient statistical reporting or incompatible outcome formats. All 13 quantitatively eligible studies were included in at least one of the two meta-analyses. This structured allocation explains the differing study counts reported across tables, text, and figures. The full selection process is illustrated in Figure 7 (PRISMA flow diagram). Title/abstract screening and full-text assessment were conducted independently by two reviewers (S.K. and W.K.). Disagreements were resolved through discussion and consensus, with arbitration by a third reviewer when necessary. Formal inter-reviewer reliability statistics were not calculated, as consensus-based resolution was applied throughout the screening process. Reasons for exclusion at the full-text stage corresponded directly to the predefined eligibility criteria and are summarized in the PRISMA flow diagram.

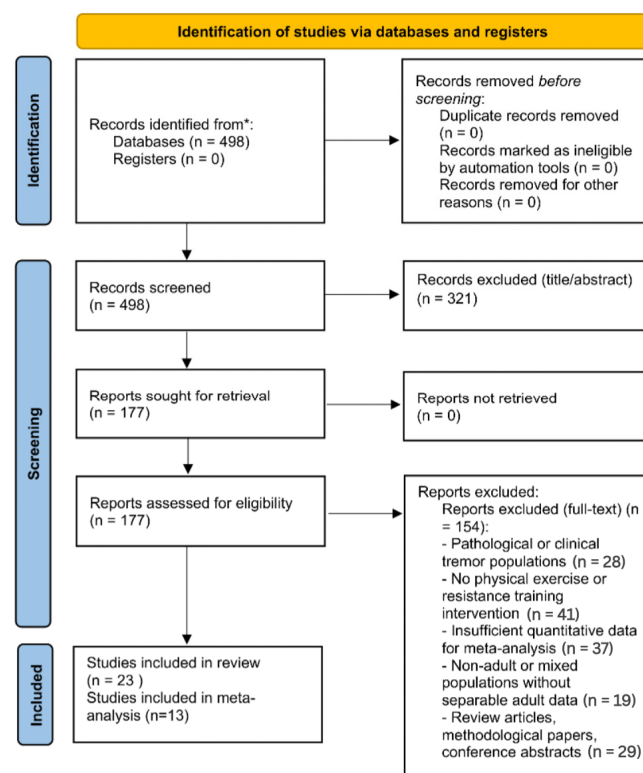


Figure 7. PRISMA 2020 flow diagram. Searches were conducted from database inception to 15 September 2025 (date last searched). Numbers shown reflect studies included at each screening stage; not all studies included after full-text screening contributed quantitative data suitable for meta-analysis. * Databases searched included PubMed, Scopus, Web of Science, and SPORTDiscus.

4.7. Data Extraction

Data were extracted independently by two reviewers using a standardized extraction form. Extracted fields included:

- Author(s), publication year, and study design.
- Sample size and participant characteristics.
- Exercise modality and intervention duration.
- Tremor measurement method (EMG, accelerometry, inertial sensors).
- Outcome metrics (RMS amplitude), with pre- and post-intervention means, SDs, and group sizes.
- Timing of tremor assessment (immediate post-exercise).

All extracted data and study identifiers (authors, publication year, and outcomes) were cross-checked across tables, figures, and analyses during revision to ensure internal consistency. When multiple post-exercise time points were reported within a study, the time point closest to exercise cessation was extracted for quantitative synthesis, as it best reflected acute effects. Additional time points were not pooled to avoid within-study dependency. When multiple measures of physiological tremor amplitude were available, magnitude-based metrics (e.g., RMS or peak power) were prioritized. If multiple analyses of the same outcome were reported, values derived from the primary analysis specified by the authors were extracted. When required dispersion measures were not directly reported, standard deviations were derived from standard errors, confidence intervals, or exact *p*-values using established statistical formulas. The overall process of study identification, screening, eligibility assessment, and inclusion is illustrated in Figure 1 (PRISMA flow diagram). All extraction decisions followed predefined rules applied consistently across studies to ensure comparability and avoid selective outcome reporting.

4.8. Risk-of-Bias Assessment

Risk-of-bias was assessed using a domain-based approach tailored to study design. For randomized controlled trials, the Cochrane RoB 2.0 framework was applied. For non-randomized and quasi-experimental pre–post studies, conceptually equivalent bias domains were evaluated (e.g., selection bias, measurement bias, missing outcome data/attrition, and selective reporting) to provide an overall methodological risk profile across the evidence base. Figures 6 and 7 present an aggregated risk-of-bias summary across all included studies. Risk-of-bias was assessed for internal interpretive purposes in randomized controlled trials using the Cochrane RoB 2.0 tool. Overall risk-of-bias in randomized trials was predominantly rated as moderate, with concerns most frequently arising in domains related to deviations from intended interventions and measurement of outcomes. Risk-of-bias was evaluated using a domain-based approach across all included studies. RoB 2.0 was applied to randomized controlled trials, whereas conceptually equivalent bias criteria were used for non-randomized pre–post studies; Figures 6 and 7 summarize the overall risk profile across the evidence base. For randomized trials, judgments were informed by RoB 2.0 principles; for non-randomized pre–post studies, conceptually equivalent criteria were applied.

4.9. Statistical Analysis

Meta-analyses were conducted using random-effects models with the DerSimonian–Laird estimator to account for expected clinical and methodological heterogeneity. The primary effect size was Hedges' *g*, reflecting standardized pre–post changes in tremor amplitude. For within-group pre–post comparisons, effect sizes were calculated for dependent samples based on change scores (post–pre). Standard deviations were derived from reported change-score variability where available; when not reported, they were

calculated from pre- and post-intervention standard deviations assuming a conservative pre-post correlation ($r = 0.5$), consistent with the recommendations of Morris and DeShon (2002) [34] for meta-analyses of repeated-measures designs. To acknowledge the potential influence of this assumption, we considered plausible correlation values ($r = 0.3$ – 0.7), as recommended by Morris and DeShon (2002) [34]. Because within-subject correlations were not consistently reported, sensitivity to r was addressed qualitatively and pooled findings were interpreted primarily in terms of effect direction.

Separate analyses were performed for all exercise interventions (research question 1; $n = 11$ studies) and for resistance-based exercise protocols only (research question 2; $n = 8$ studies). Heterogeneity was quantified using Cochran's Q test ($p < 0.10$), the I^2 statistic (interpreted as low [25%], moderate [50%], or high [$\geq 75\%$] heterogeneity), and Tau^2 as an estimate of between-study variance. Publication bias was evaluated using funnel plots. Egger's regression test was considered when at least 10 studies were available; however, due to the borderline number of studies ($n = 10$ – 11) and the known instability of the test in small samples, Egger's test was not applied, and publication bias was assessed descriptively.

To support transparent reporting and study characterization, we extracted descriptive protocol characteristics, including the exercise type (e.g., isometric, eccentric), measurement modality (e.g., EMG, accelerometry/IMU), muscle group assessed (upper vs. lower limb), and timing of post-exercise assessment (e.g., immediate vs. delayed). No formal subgroup analyses or meta-regression were performed. These characteristics were used for descriptive classification and narrative interpretation only, and any apparent patterns across categories should be considered hypothesis-generating.

- All analyses were conducted using Review Manager 5.4 and Comprehensive Meta-Analysis v4, with $p < 0.05$ as the threshold for statistical significance. If required statistical parameters were missing or unclear, standard deviations were derived from standard errors, confidence intervals, or p -values when possible, using established formulas. Studies for which effect sizes could not be reliably calculated were excluded from quantitative synthesis. Prespecified descriptive categories (e.g., exercise modality and measurement technique) were summarized to aid interpretation of methodological variability rather than to generate category-specific effect estimates. Statistical significance of effect sizes was defined conservatively as 95% confidence intervals that did not include zero; confidence intervals terminating at zero were interpreted as not statistically significant. Data extraction and meta-analytic computations were performed using a standardized but non-automated workflow based on the summary statistics reported in the included studies. The two meta-analyses were based on partially overlapping study sets and should be interpreted as complementary syntheses rather than statistically independent comparisons.

5. Conclusions

This meta-analysis provides the first systematic and quantitative synthesis of experimental evidence indicating that physiological tremor amplitude can be acutely modulated by physical exercise, with reductions observed across acute exercise modalities, including resistance-based exercise protocols. Across the 13 included studies, both general exertion and structured strength-based protocols were associated with a reduction in tremor amplitude in healthy adults. Reductions were reported across multiple acute protocol types (e.g., isometric and eccentric exercise protocols). These observations may reflect peripheral desynchronization and transient changes in central oscillatory drive; however, mechanistic interpretations remain theoretical and were not directly tested in the included studies. The findings underscore the relevance of physiological tremor assessment as a sensitive, non-invasive outcome measure reflecting short-term neuromuscular state under

experimental conditions. By synthesizing evidence separately for general acute exercise and acute resistance-based protocols, this study provides additional context for interpreting tremor as a functional output measure. The inclusion of studies using modern sensor modalities (e.g., surface EMG and inertial measurement units) highlights the feasibility of incorporating tremor metrics into experimental neuromuscular research protocols. Despite significant between-study heterogeneity ($I^2 > 65\%$ in both models), the directionality of effects was consistent across studies. Funnel plot inspection was exploratory and inconclusive due to the limited number of studies; therefore, publication bias cannot be ruled out. Together, these findings suggest that tremor analysis may be a promising research outcome measure in future controlled experimental studies investigating fatigue and neuromuscular function. However, given the predominance of pre–post designs, heterogeneity across protocols, and the absence of formal certainty grading (e.g., GRADE), these conclusions should be interpreted cautiously and primarily as hypothesis-generating. Future work should explore longer-term adaptations, modality-specific thresholds, and population-specific responsiveness to fully harness the diagnostic and performance potential of physiological tremor monitoring.

5.1. Practical Implications and Future Directions

These findings underscore that physiological tremor appears sensitive to short-term shifts in neural drive, muscle activation patterns, and afferent feedback. In applied contexts, tremor amplitude may inform hypothesis generation and exploratory monitoring approaches in areas such as rehabilitation, sports science, and motor learning [22], sports monitoring [4], or motor learning.

Future research should prioritize the standardization of tremor metrics across protocols, the inclusion of follow-up measures (delayed tremor recovery), multimodal monitoring (EEG–EMG–IMU), and application to high-performance and aging populations.

5.2. Limitations

Several limitations should be considered when interpreting the findings of this meta-analysis. Importantly, in the revised quantitative synthesis, only acute single-session exercise protocols were pooled, as acute fatigue effects and multi-week training adaptations reflect distinct physiological mechanisms and causal estimands. Accordingly, multi-week training interventions (≥ 2 weeks) were excluded from quantitative pooling and were summarized narratively only. Acute tremor modulation likely reflects transient changes in motor unit behavior and central drive, whereas training interventions may additionally involve adaptive processes; therefore, the pooled estimates should be interpreted as reflecting acute modulation only, primarily in terms of effect direction rather than modality-specific magnitude.

First, the number of studies eligible for quantitative synthesis was relatively small ($n = 13$), which limited the statistical power for exploratory comparisons and precluded robust subgroup modeling, and also precluded robust application of publication bias diagnostics such as Egger's regression or trim-and-fill correction. Although the directionality of effects was generally consistent, estimates of heterogeneity were moderate to high, indicating substantial variability across study designs, populations, and tremor assessment protocols.

Second, the included studies differed in the type, intensity, and acute characteristics of the exercise interventions, ranging from single-session isometric exercise bouts to eccentric and supramaximal protocols. Although multi-week training studies were identified, these were included in the qualitative synthesis only. This heterogeneity, while reflective of real-world diversity, introduces challenges in establishing modality-specific conclusions.

Furthermore, not all studies reported critical details such as the timing of tremor assessment relative to exercise cessation or the use of standardized warm-up procedures, both of which may influence tremor expression. Third, despite focusing on healthy adult populations, there was limited demographic and sex-specific stratification in the source studies. Age-related variability in tremor responsiveness and potential sex differences in fatigue recovery or neuromuscular control could not be fully explored. Additionally, some included studies lacked complete statistical data, requiring author contact or inference of variance estimates, which, although methodologically justified, introduces a degree of imprecision. Moreover, effect size estimation in pre–post designs required assuming within-subject correlations when SD_change was unavailable; although sensitivity analyses supported robustness across plausible r values, this remains a source of uncertainty. Finally, the majority of studies employed only one measurement modality (e.g., surface EMG or inertial sensors) and rarely validated tremor changes across multiple channels of neuromechanical data. The absence of neurophysiological correlates such as EEG or cortico-muscular coherence restricts deeper mechanistic interpretation. The certainty of evidence was not formally assessed using a GRADE or equivalent framework; therefore, conclusions should be interpreted with appropriate caution. Although this review adheres to PRISMA 2020 reporting standards, no fully reproducible analytic pipeline or publicly accessible code repository was implemented. Extracted datasets and analytic outputs are available via Zenodo (DOI: 10.5281/zenodo.15487242), supporting transparency and computational reproducibility; however, the analytic workflow was not fully automated. These limitations highlight the need for greater standardization in tremor research methodology and more comprehensive reporting of exercise protocols and neuromechanical outcomes.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/physiologia6010011/s1>, Supplementary Table S1: Database-specific search strategies; Supplementary Table S2: PRISMA 2020 Checklist.

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Abbreviations

The following abbreviations are used in this manuscript:

RMS	Root Mean Square
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
EMG	Electromyography
IMU	Inertial Measurement Unit
SD	Standard Deviation
CI	Confidence Interval
SE	Standard Error
RoB 2.0	Risk-of-Bias 2.0 (Cochrane Risk of Bias Tool, Version 2)
JBI	Joanna Briggs Institute
EEG	Electroencephalography

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